

Machine Learning (CSCI - 4465)

Lab Assignment One

October 4, 2025

Instruction: In this assignment, you will implement Linear Regression with the Gradient Descent algorithm and the Normal Equations. Please attach your code and figures with your submission.

1. Let us assume that for a regression problem, we have a training set consisting of 200 examples of (x, y) pair and that we only have one feature X (which could be the size of a house, for instance). Also, let us assume that the target variable Y (e.g., the price of a house) is a quadratic function of the input feature X of the form:

$$y = \theta_0 + \theta_1.x + \theta_2.x^2,$$

where you can assume θ_0 , θ_1 and θ_2 are uniformly distributed over the interval $[0, 1]$ and $X \sim \mathcal{N}(0, 1)$.

Implement the following:

- Generate the dataset with 200 random observations.
- Implement Gradient Descent to obtain estimate of the parameter vector $\theta = [\theta_0, \theta_1, \theta_2]^T$. You can assume that the algorithm has converged when the cost function $J(\theta)$ calculated in successive iterations of the algorithm is not changing above 10^{-4} , i.e.: $|J^{(t+1)}(\theta) - J^{(t)}(\theta)| < 10^{-4}$, where t denotes the number of iterations of the algorithm. Show the plot showing the final quadratic fit to the training dataset.
- Also solve for θ directly using Normal Equations and show the plot of the final quadratic fit to the dataset.
- Compare the two estimates of θ obtained from Gradient Descent and from the Normal Equations. Check if you get a zero vector when you subtract the two estimates of θ , i.e., whether $\theta_{GD} - \theta_{NE} = \vec{0}$.